TF

TaskFlow

COMPLETE ILLUSTRATED PRODUCT GUIDE

Every workflow. Every role. Every screen.

Administrator assignment, status-only assignee controls, private personal to-dos, dashboards, Kanban, reminders, data tools, security, setup, and responsive behavior.

ADMINISTRATOR

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Generated 21 October 2026 | 19 application screenshots

DOCUMENT MAP

How to use this guide

A complete visual and technical reference for TaskFlow.

01

Access and identity

Sign-in, account creation, secure sessions, and role recognition.

02

Administrator workspace

Assignment dashboard, task lifecycle, filters, Kanban, and data tools.

03

Personal workspace

Private to-dos kept completely separate from administrator-assigned work.

04

Normal-user workspace

Read-only assigned-task details with status as the only permitted change.

05

Responsive experience

Desktop, mobile dashboard, navigation drawer, forms, and mobile authentication.

06

Technical reference

Permissions, email timing, API, MySQL, security, setup, and verification.

REFERENCE

Role and permission model

The API enforces these permissions even when a user bypasses the interface.

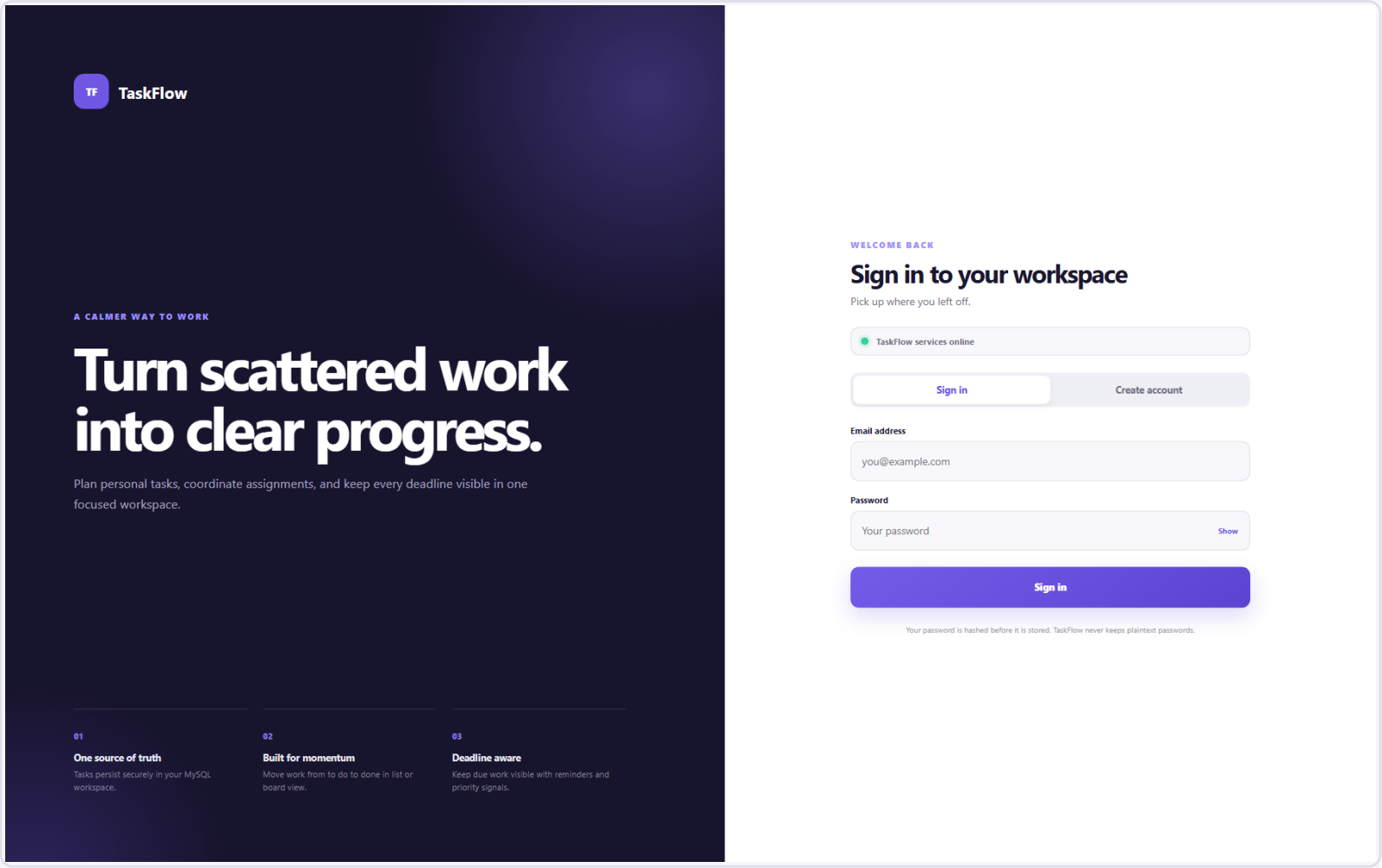
ACTION	ADMIN - ASSIGNED WORK	ASSIGNEE - ASSIGNED WORK	PERSONAL OWNER
View task	Yes	Yes, when assigned	Yes, own tasks only
Create task	Yes, for any registered user	No assigned-work creation	Yes, private personal task
Edit title/description	Yes	No	Yes
Change priority/deadline	Yes	No	Yes
Reassign task	Yes	No	Not applicable
Change status	Yes	Yes - only permitted change	Yes
Delete task	Yes	No	Yes
View user directory	Yes	No	No
Trigger development reminder	Yes	No	Yes for own personal task

Normal users can change an assigned task only between To do, In progress, and Done. Attempts to modify any other field return HTTP 403; deletion returns no accessible task.

ILLUSTRATED FEATURE

Secure sign-in

Entry point for returning users.

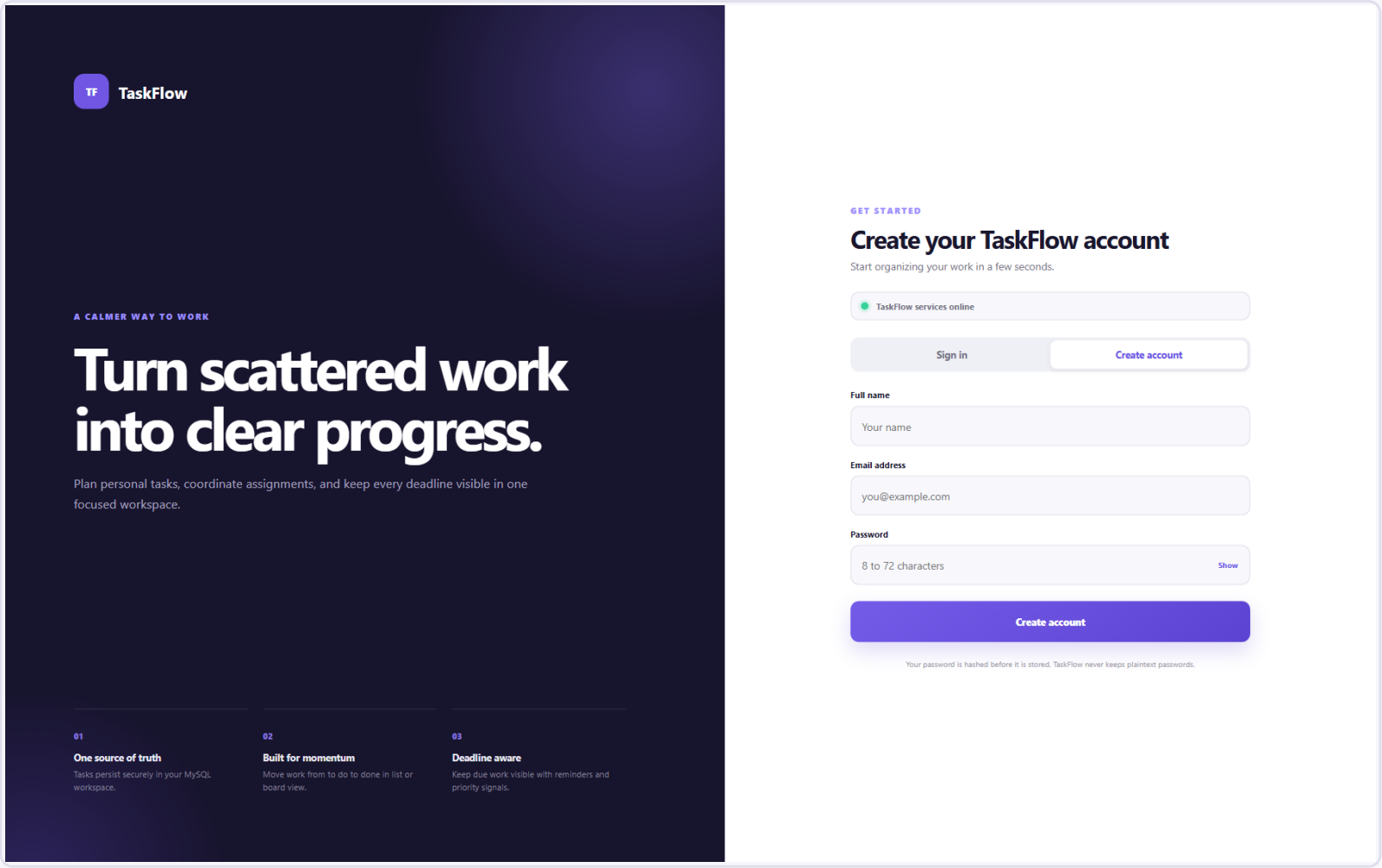


- Email/password authentication backed by bcrypt and JWT.
- Live service badge confirms API and MySQL availability.
- Session restoration keeps signed-in users in their workspace.

ILLUSTRATED FEATURE

Account registration

New users create their own TaskFlow identity.

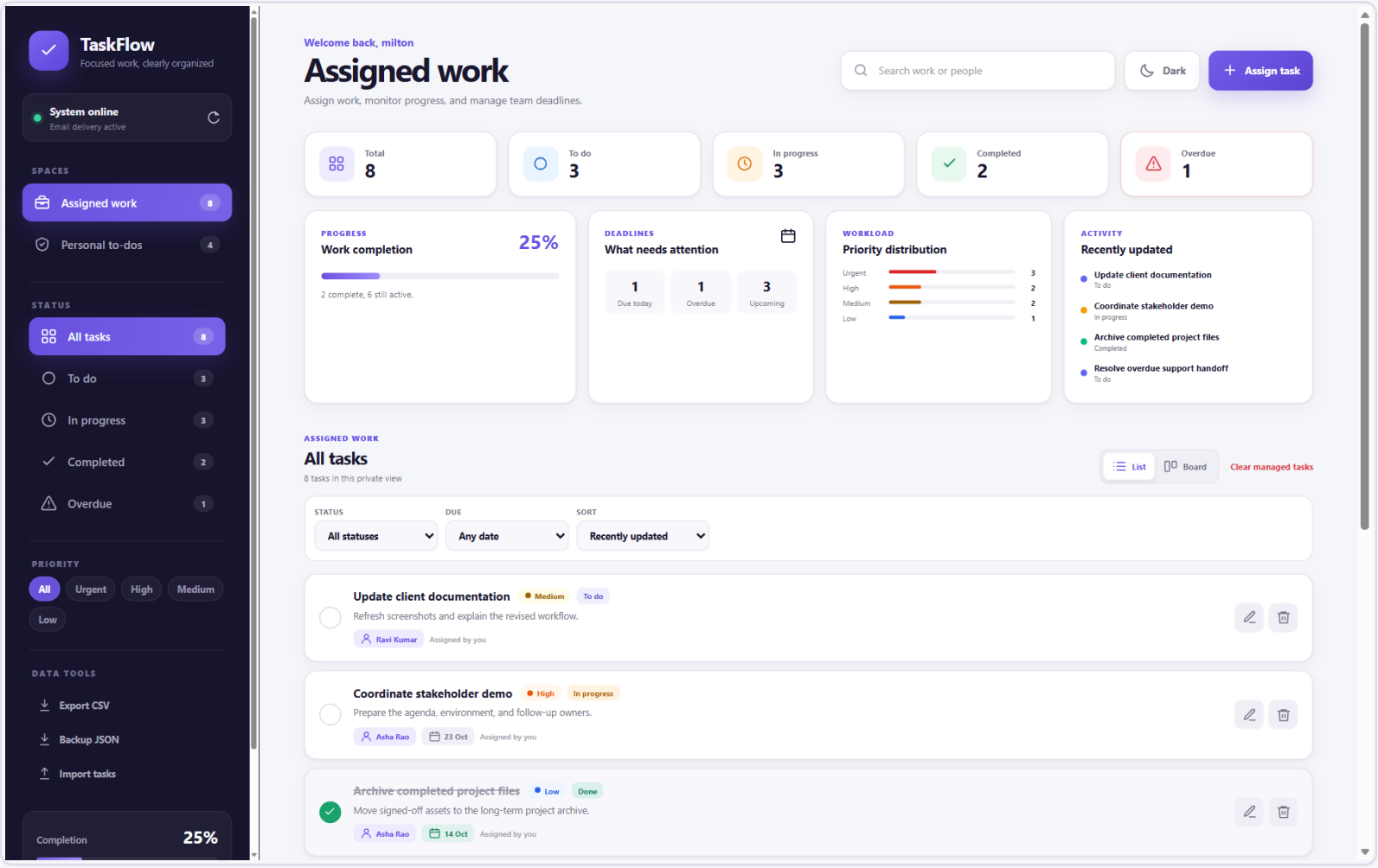


- Validated full name, email, and 8-72 character password.
- Passwords are hashed before database storage.
- The configured administrator email is automatically promoted to admin.

ILLUSTRATED FEATURE

Admin assigned-work dashboard

A real-time overview of delegated work.

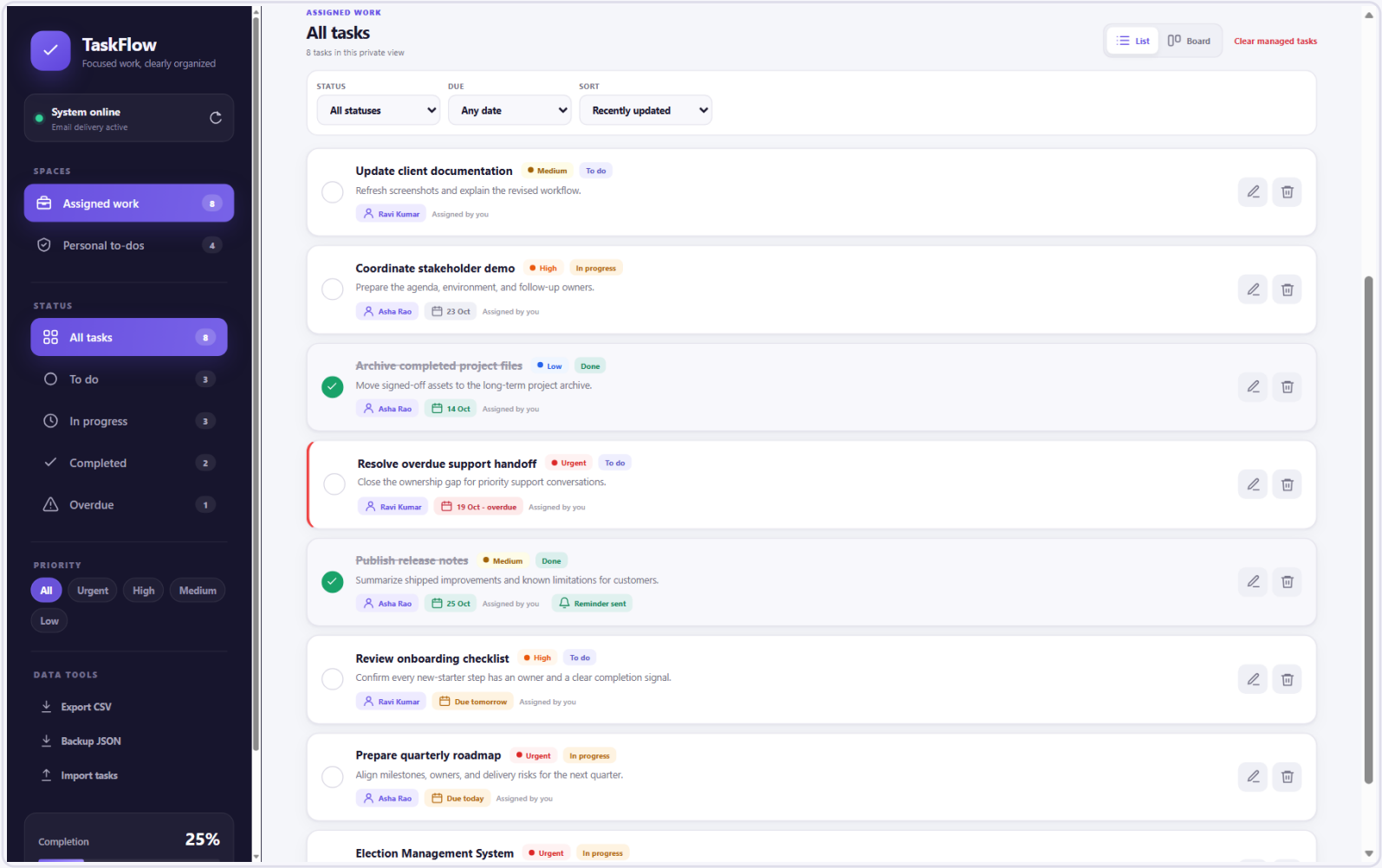


- Totals for To do, In progress, Completed, and Overdue.
- Completion rate, due-date attention, priority distribution, and recent activity.
- Admin-only assignment button and full registered-user directory.

ILLUSTRATED FEATURE

Admin task list and data tools

Detailed management view for assigned work.

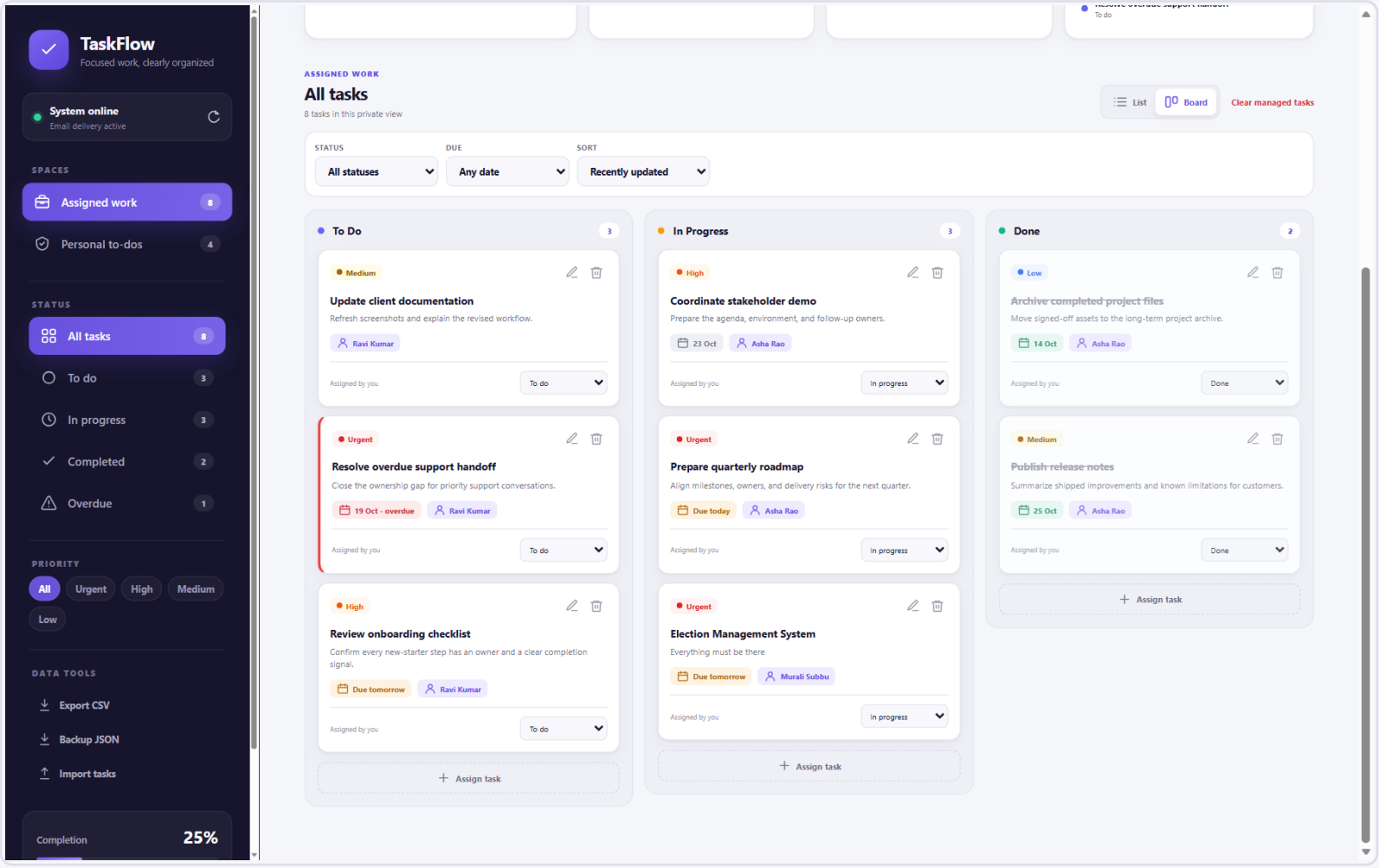


- Edit and delete controls appear only for the admin.
- CSV export, JSON backup, and validated import.
- Status, deadline, assignee, creator, reminder, and priority metadata.

ILLUSTRATED FEATURE

Kanban workflow

The same persisted tasks organized by workflow stage.

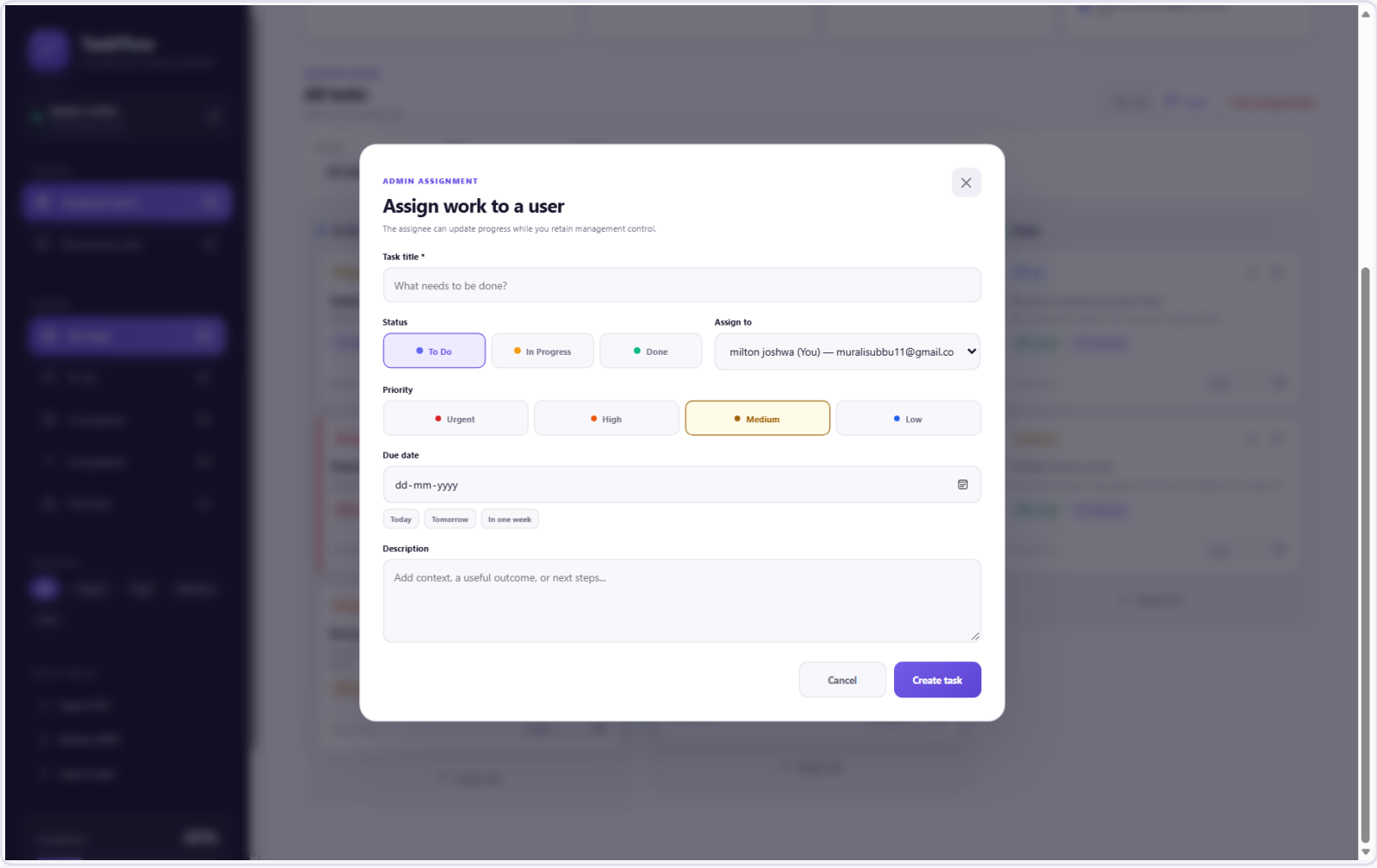


- To Do, In Progress, and Done columns.
- Drag-and-drop status changes with server persistence.
- Column-level task creation and complete admin controls.

ILLUSTRATED FEATURE

Assign work to a registered user

Administrator-only assignment form.

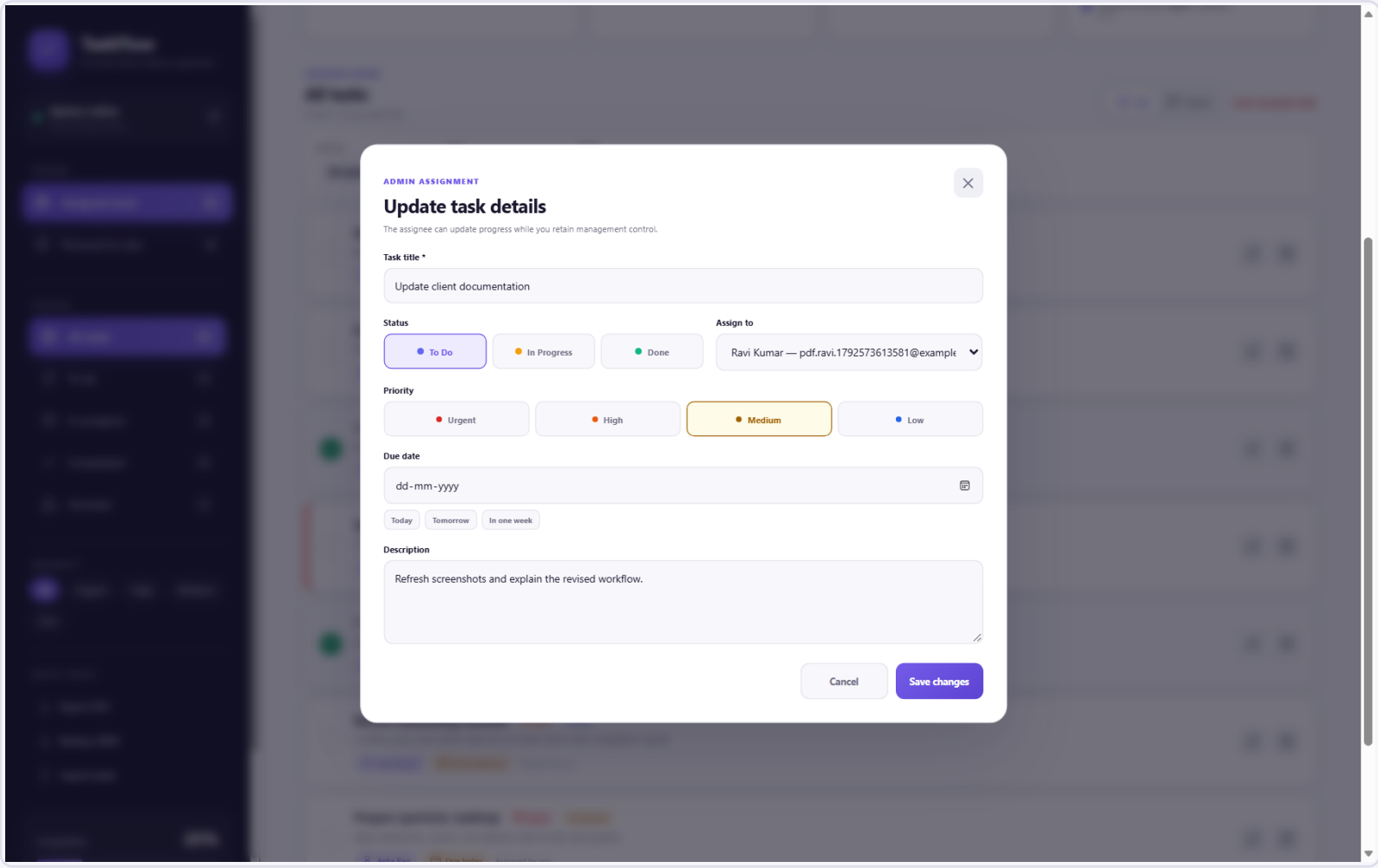


- Choose assignee, status, priority, due date, and description.
- Quick due-date shortcuts for today, tomorrow, and one week.
- Successful creation can send an immediate assignment email.

ILLUSTRATED FEATURE

Edit and reassign work

The admin retains management control after delegation.

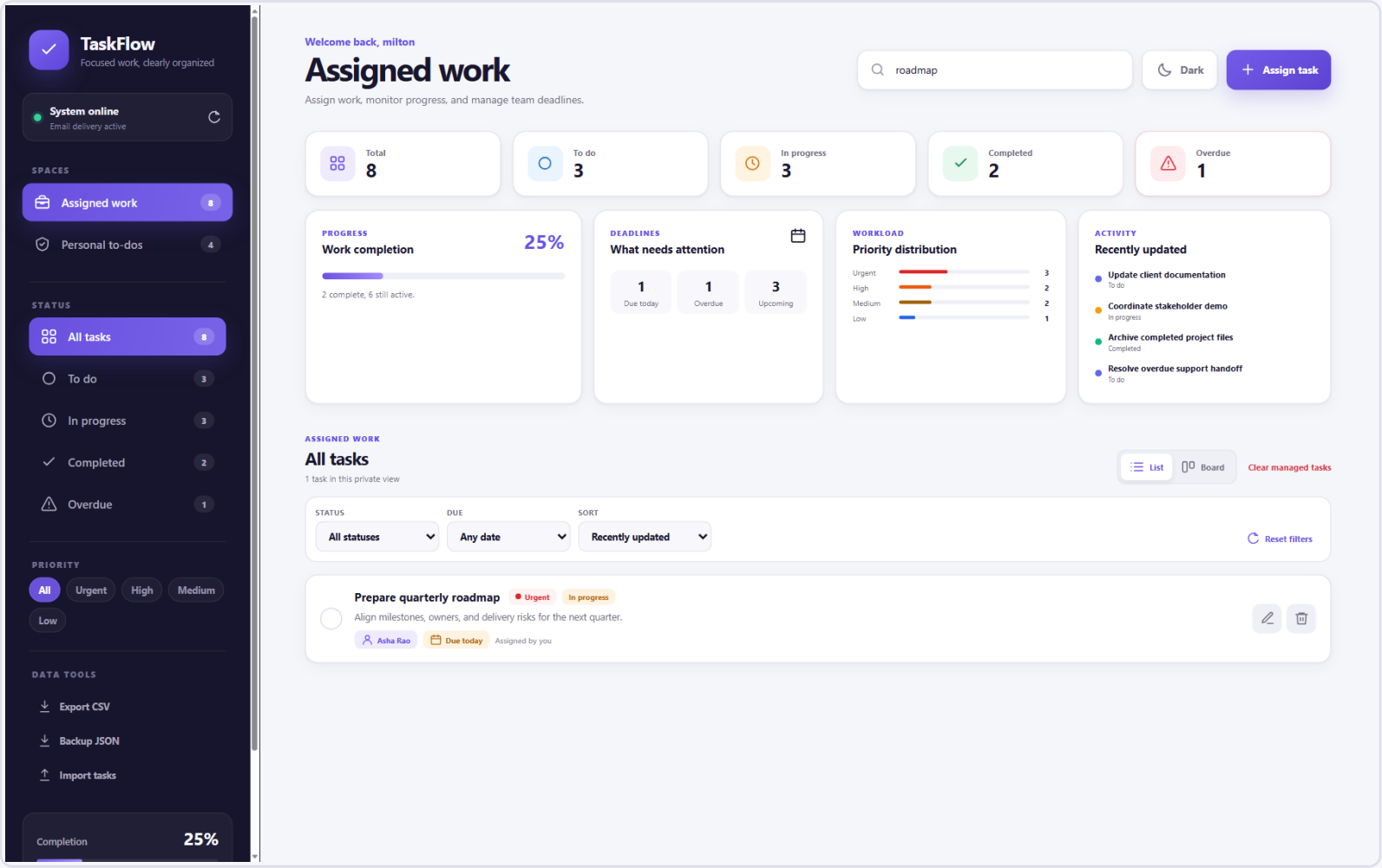


- Update every task detail and reassign to another registered user.
- Reassignment can send an immediate notification to the new assignee.
- Reminder eligibility resets safely when deadline or assignee changes.

ILLUSTRATED FEATURE

Search, filter, and sorting

Focused navigation through larger task sets.

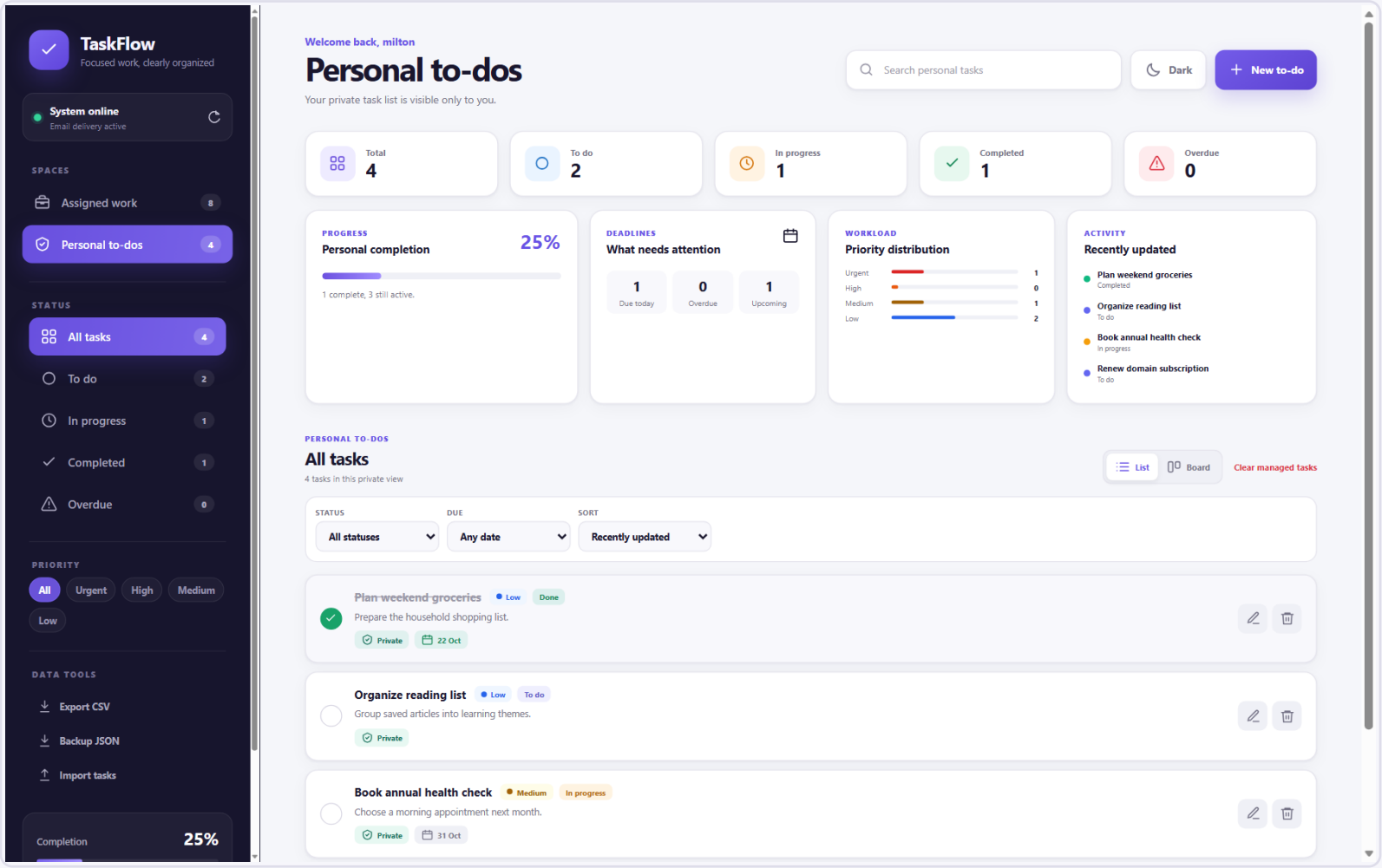


- Search title, description, assignee, or creator.
- Combine status, priority, and due-date filters.
- Sort by update time, creation time, deadline, priority, or title.

ILLUSTRATED FEATURE

Separate personal workspace

Administrator personal to-dos never mix with delegated work.

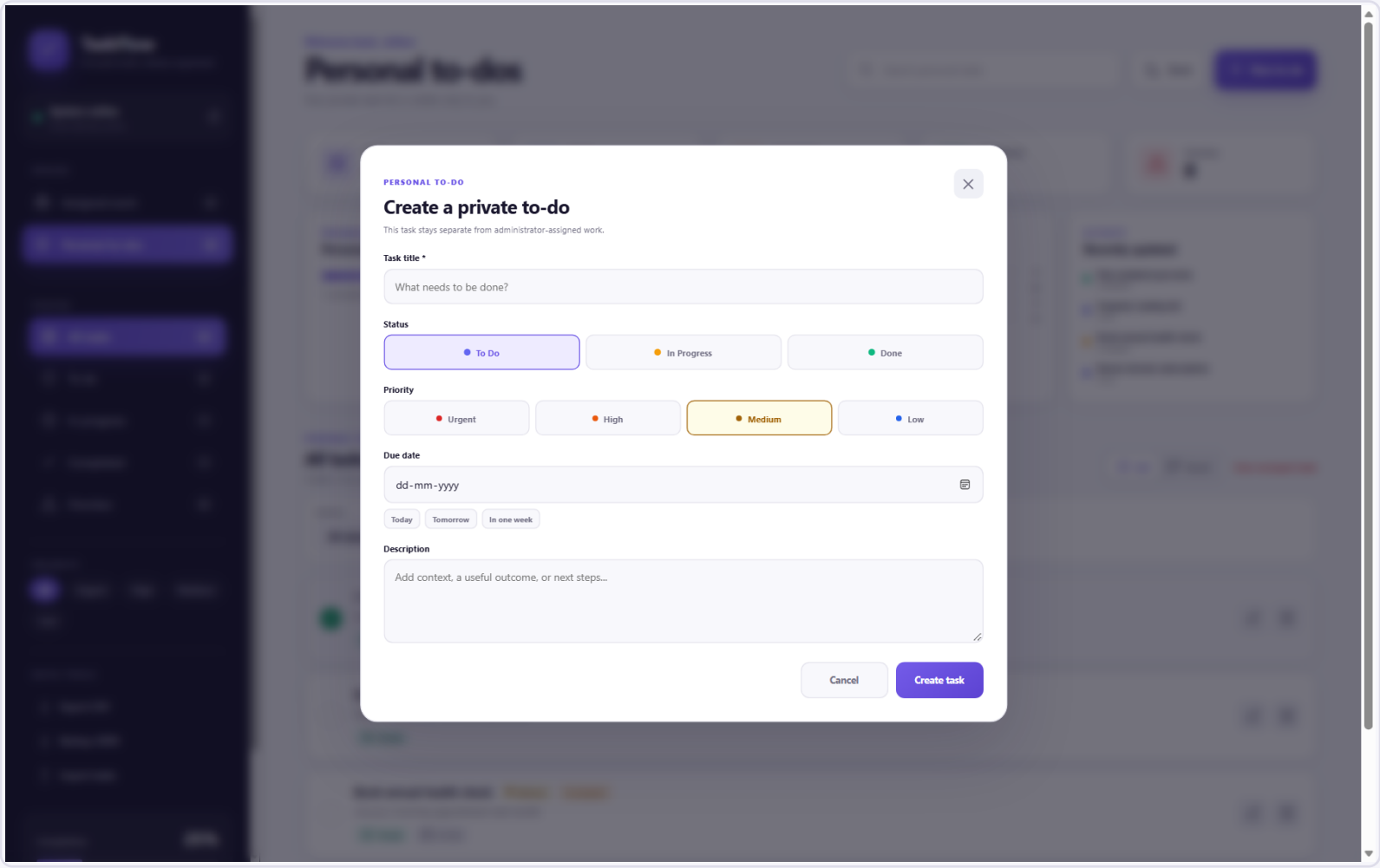


- Independent counts, dashboard analytics, filters, exports, and deletion.
- Personal items are visible only to their owner.
- Switching spaces resets filters to avoid confusing cross-space results.

ILLUSTRATED FEATURE

Create a private personal to-do

A focused form without assignment controls.

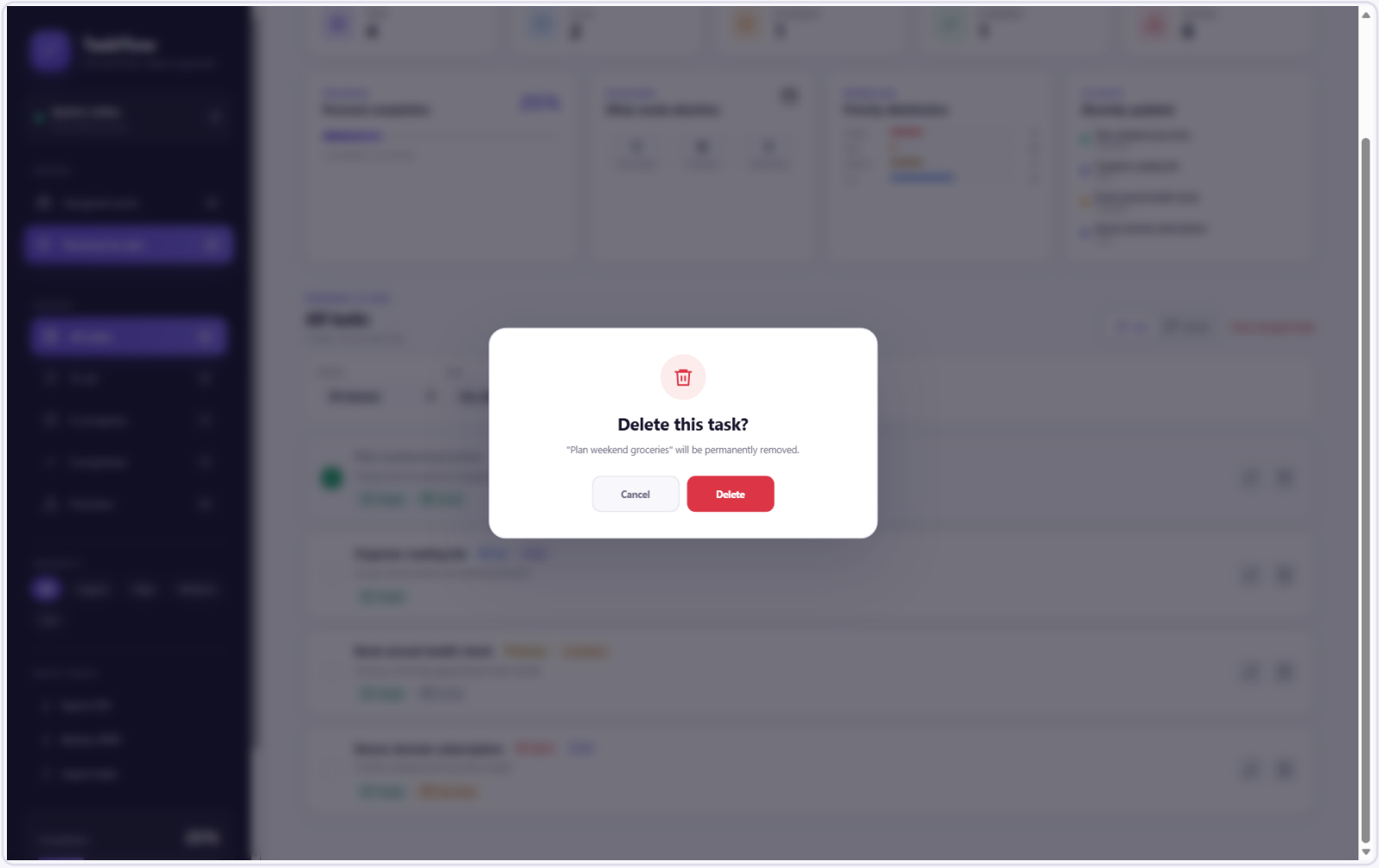


- Title, workflow status, priority, due date, and description.
- No user directory or assignee picker is exposed.
- The API forces personal ownership to the signed-in account.

ILLUSTRATED FEATURE

Protected destructive actions

Deletion requires deliberate confirmation.

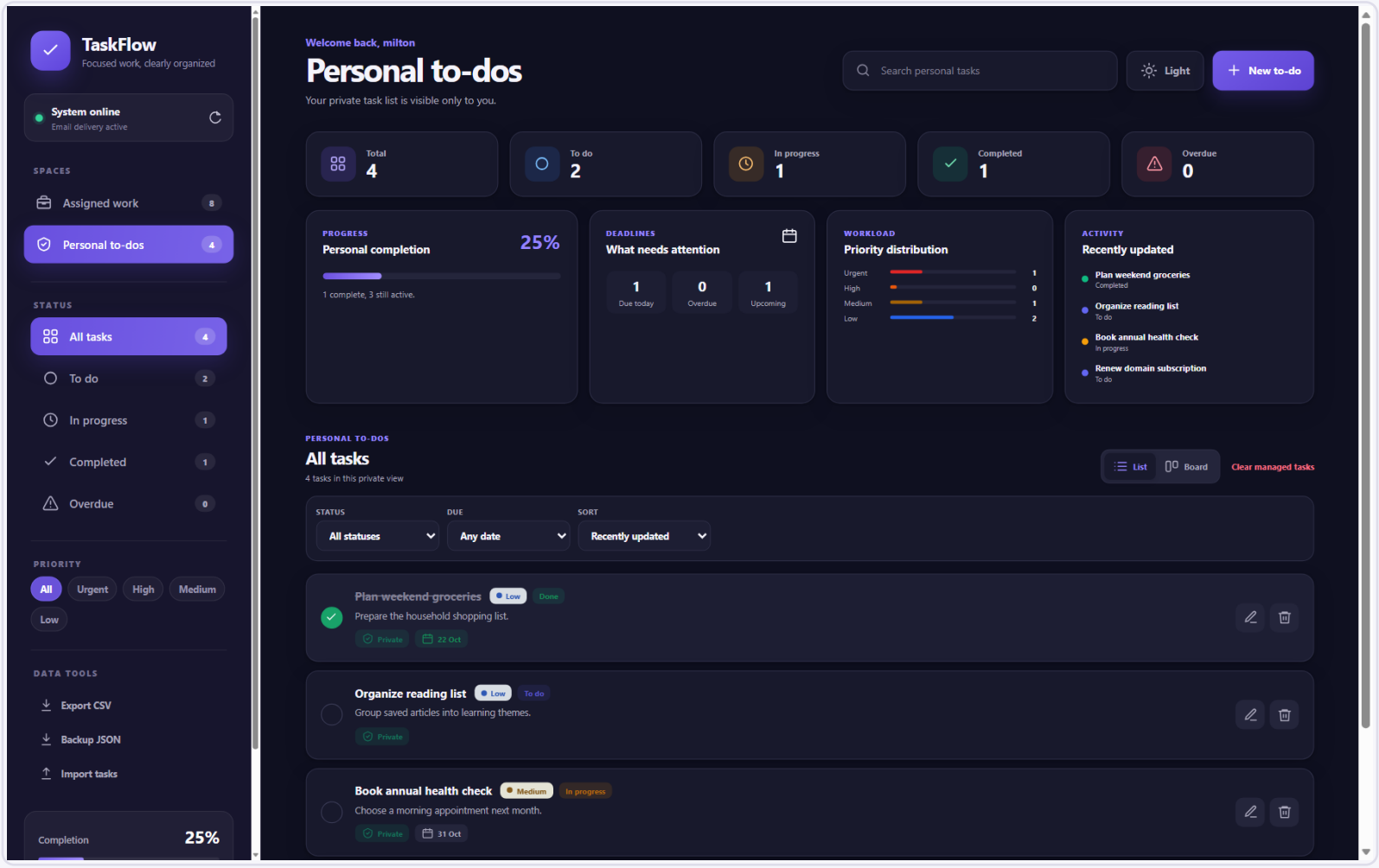


- The dialog names the exact task being removed.
- Bulk clearing is scoped to the current personal or assigned space.
- The opposite workspace remains untouched.

ILLUSTRATED FEATURE

Dark theme

A persistent alternate visual mode.

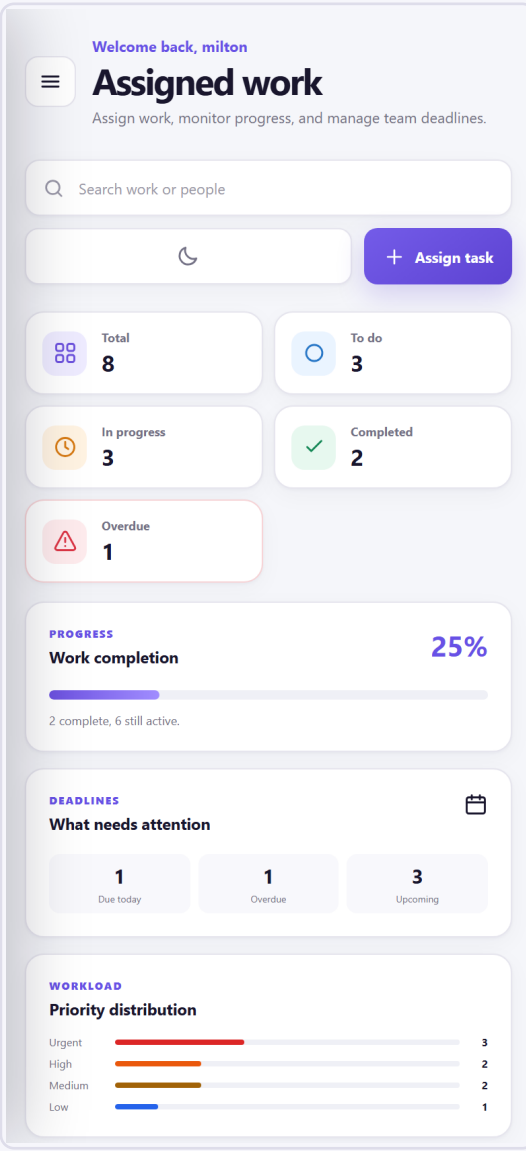


- Theme preference is stored in the browser.
- All cards, forms, states, and navigation remain legible.
- Theme selection does not affect server-side data.

ILLUSTRATED FEATURE

Responsive mobile dashboard

TaskFlow adapts dashboard content to a narrow screen.



WHAT THIS SCREEN DEMONSTRATES

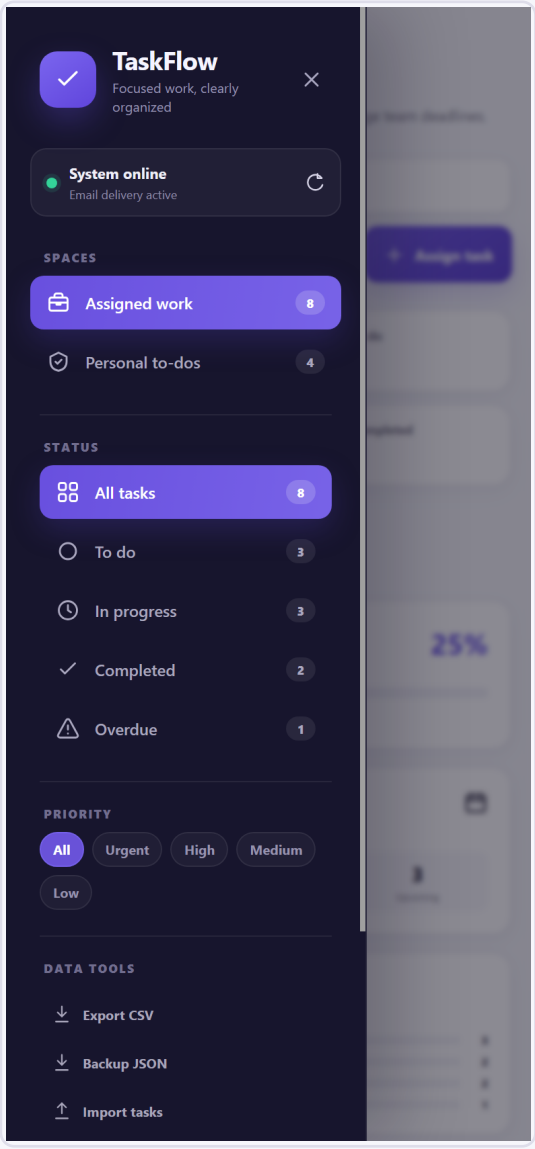
- Summary cards reflow without horizontal page overflow.
- Primary assignment action remains prominent for the admin.
- Analytics stack vertically for touch-friendly reading.

Captured from: 13-mobile-dashboard.png

ILLUSTRATED FEATURE

Mobile navigation drawer

All workspace and filter controls remain accessible on mobile.



WHAT THIS SCREEN DEMONSTRATES

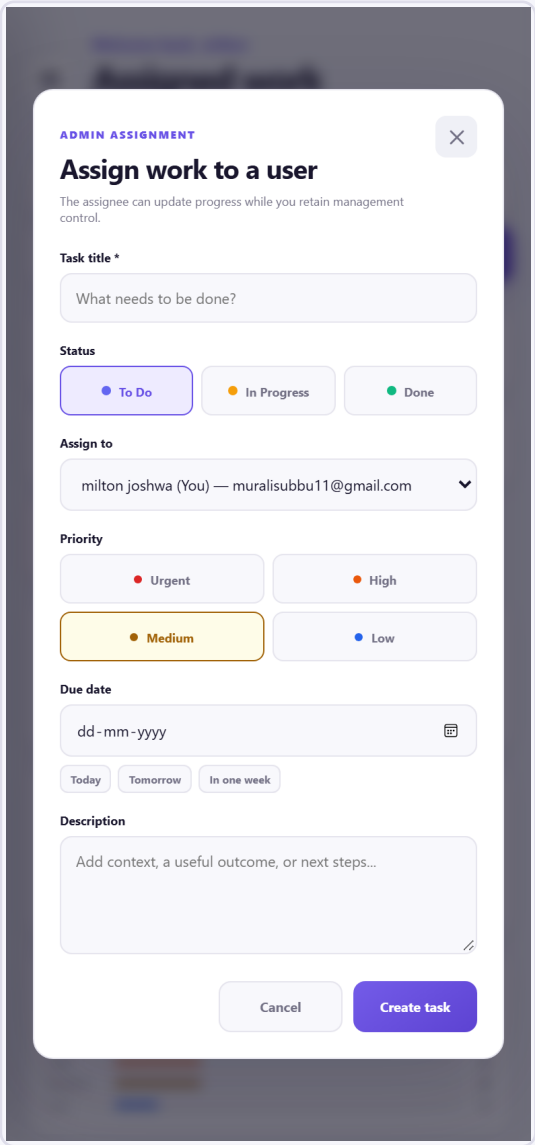
- Drawer separates Assigned work and Personal to-dos.
- Status, priority, and data tools remain available.
- Backdrop and close controls support predictable navigation.

Captured from: 14-mobile-navigation.png

ILLUSTRATED FEATURE

Mobile assignment form

The full admin form remains usable on a phone-sized viewport.



WHAT THIS SCREEN DEMONSTRATES

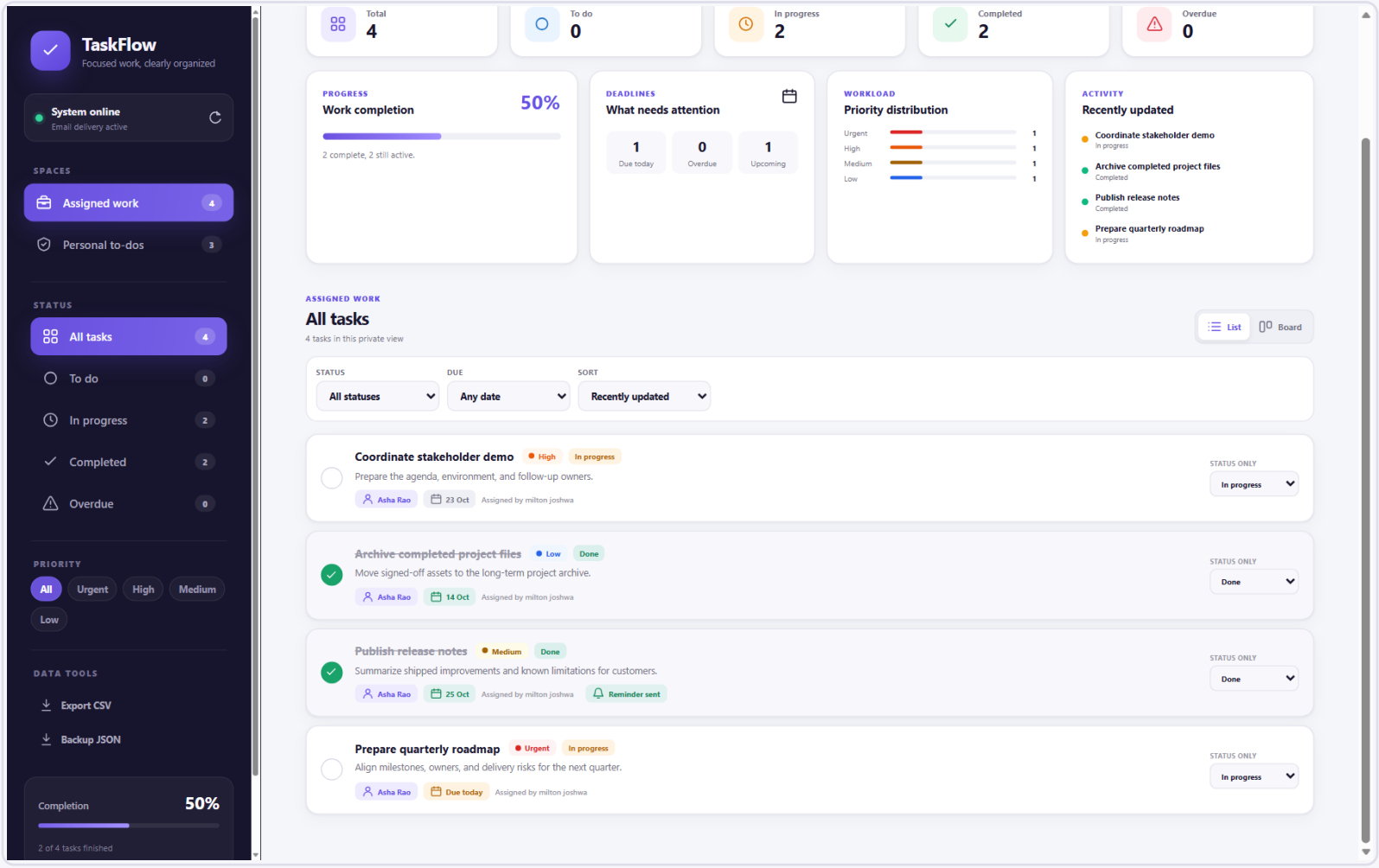
- Scrollable modal stays inside the viewport.
- Touch targets remain large enough for status and priority selection.
- Assignee and due-date controls preserve desktop capability.

Captured from: 15-mobile-task-modal.png

ILLUSTRATED FEATURE

Normal user: status-only assigned work

Assigned details are read-only; status is the sole action.

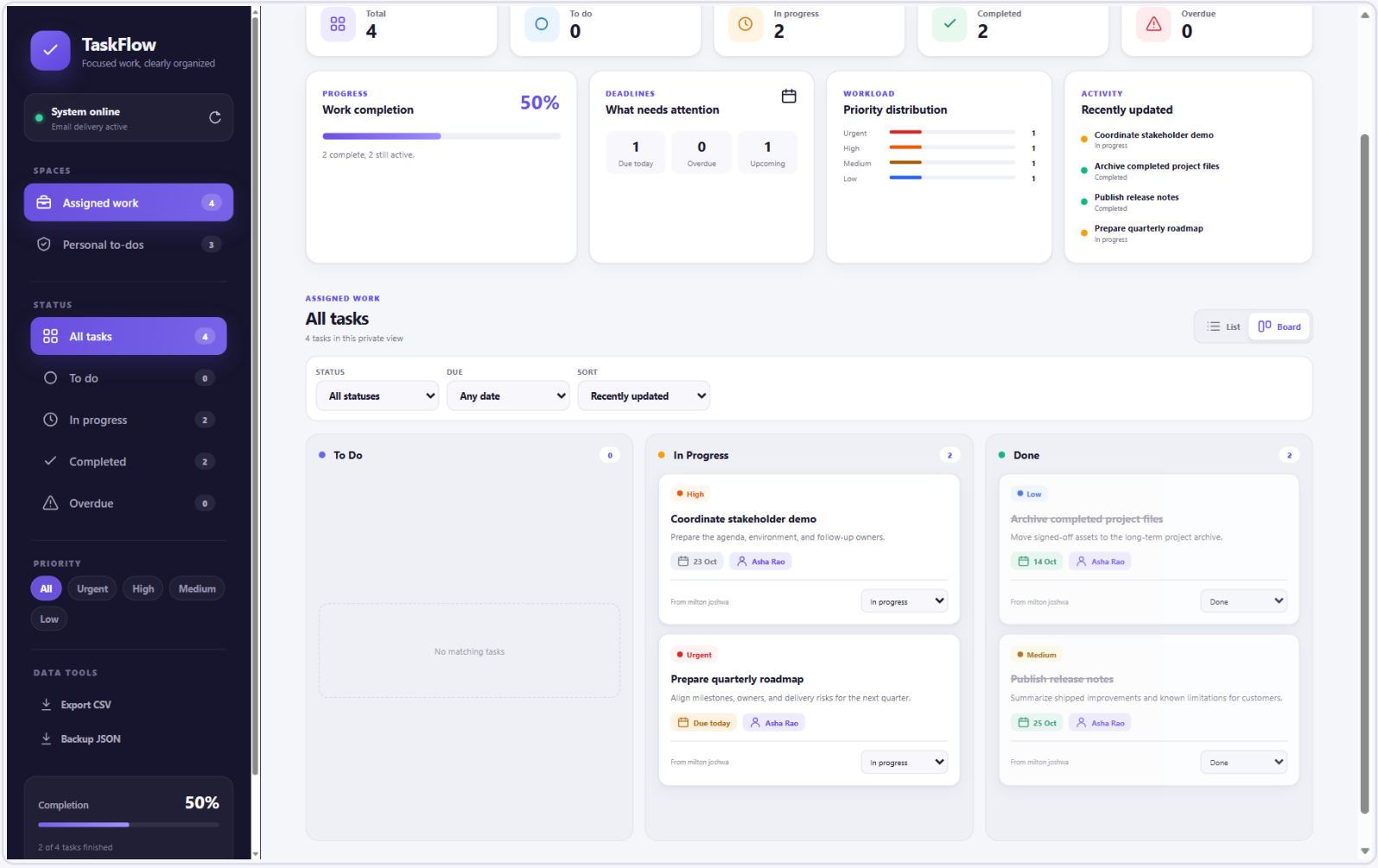


- No edit, delete, reassign, or create-assignment controls.
- Each card explicitly displays a STATUS ONLY selector.
- The backend rejects title, description, priority, deadline, assignee, type, and deletion changes.

ILLUSTRATED FEATURE

Normal user Kanban

Assignees can move work through its workflow, but cannot edit its content.

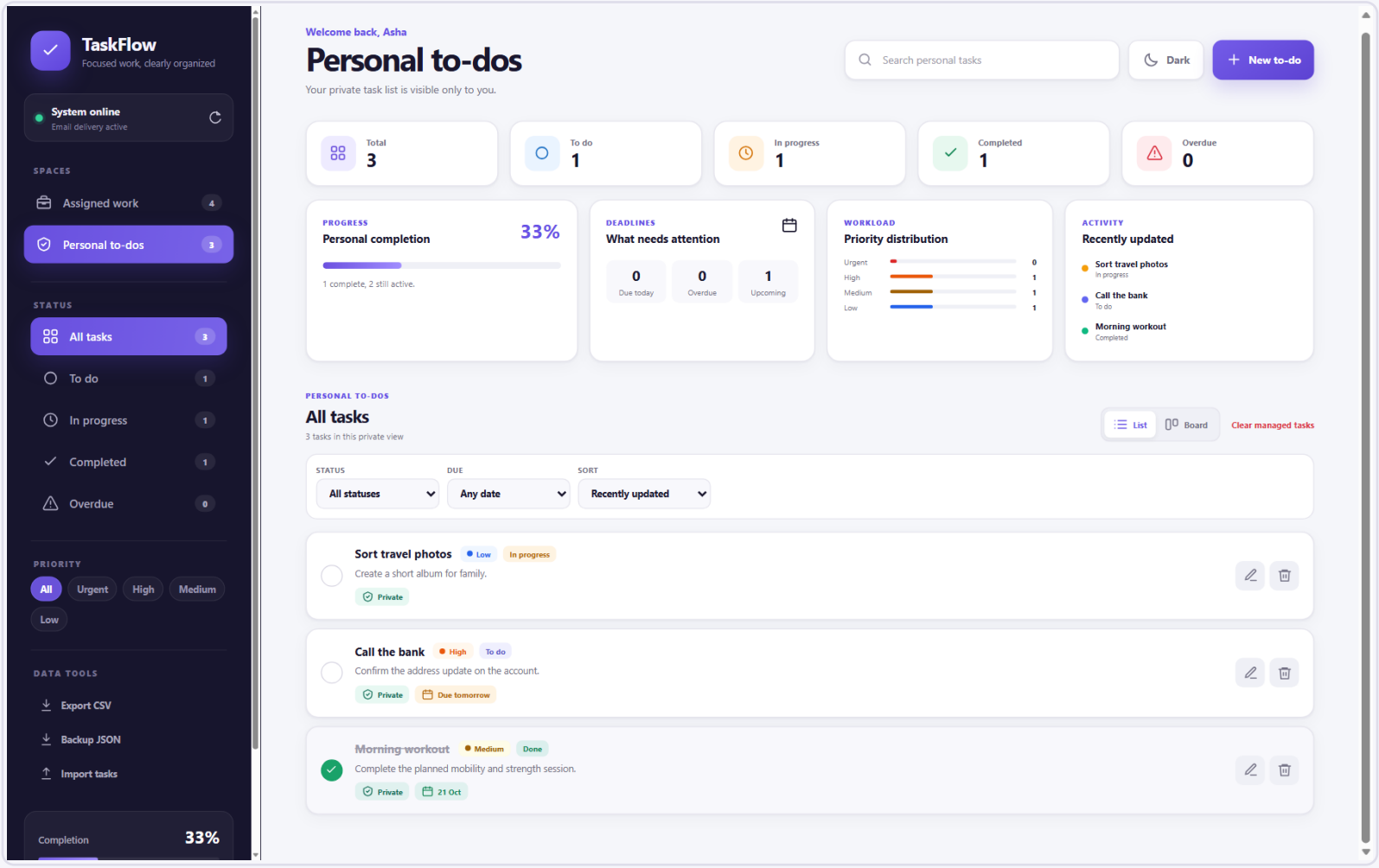


- Drag/drop and status dropdowns change workflow status only.
- Task details, priority, deadline, and assignment remain admin-owned.
- Only tasks assigned to the signed-in user are visible.

ILLUSTRATED FEATURE

Normal user personal to-dos

Every registered user has a private productivity space.

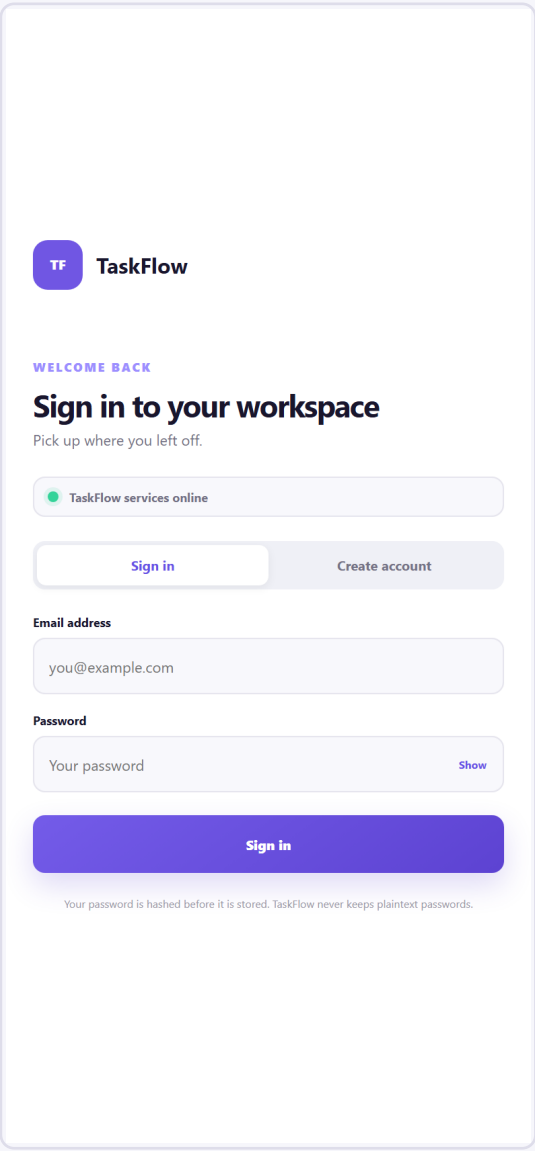


- Users can fully manage their own personal items.
- Personal items remain separate from admin-assigned work.
- The administrator cannot view another user's personal list.

ILLUSTRATED FEATURE

Mobile authentication

The authentication flow remains clear on small devices.



WHAT THIS SCREEN DEMONSTRATES

- Compact branding and accessible form controls.
- Sign-in/create-account tabs remain available.
- Service status and password visibility controls are retained.

Captured from: 19-mobile-sign-in.png

TECHNICAL REFERENCE

Complete feature catalog

Functional coverage implemented in the current TaskFlow build.

Identity and access

- Register, sign in, session restore, sign out
- bcrypt password hashing; JWT issuer and expiry validation
- Single configurable admin email; live role lookup on every authenticated request

Task lifecycle

- Create, view, edit, status change, reassign, delete, scoped bulk clear
- Statuses: To do, In progress, Done
- Priorities: Low, Medium, High, Urgent; optional due date and description

Workspace separation

- Admin-assigned work with controlled delegation
- Private personal to-do list for every account
- Separate analytics, filters, import/export, and deletion for each space

Productivity views

- List and Kanban representations of the same server data
- Dashboard counts, completion percentage, deadline metrics, priorities, recent activity
- Search, compound filters, sorting, overdue highlighting, reminder indicators

Data portability

- CSV export protected against spreadsheet formula injection
- JSON backup and validated import
- Maximum 200 imported tasks per operation; server validation still applies

Experience

- Light/dark themes
- Responsive desktop and mobile navigation
- Loading, empty, error, confirmation, and toast states; keyboard Escape support

OPERATIONS

Email notifications and reminder timing

SMTP mode was active during this documentation capture.

01

Admin creates assigned work

An assignment email is attempted immediately for the selected registered user.

02

Admin reassigns work

The new assignee receives an immediate assignment notification.

03

Due-date reminder

The daily scheduler checks at 09:00 Asia/Kolkata by default.

04

Task due today is created/updated

An immediate reminder check runs after the save so urgent work is not missed.

05

Personal deadline

No assignment email is sent; the owner can receive a due-date reminder when a due date exists.

06

Delivery failure

Failed SMTP delivery is logged and remains eligible for retry; simulated mail is never marked delivered.

Requirements for real delivery: the API must be running, MySQL must be reachable, the recipient must have a valid registered email, and SMTP_HOST / SMTP_PORT / SMTP_USER / SMTP_PASS / SMTP_FROM must be valid. Gmail requires an app password rather than the normal account password.

REFERENCE

API and data model reference

The React client uses /api through the Vite development proxy; all writes are persisted in MySQL.

AREA	ENDPOINT / TABLE	PURPOSE
Health	GET /api/health	Reports database, email mode, timezone, and development-tool state
Authentication	POST /api/auth/register; /login	Create account and issue signed JWT
Session	GET /api/auth/me	Refresh current role and profile from MySQL
Users	GET /api/users	Admin-only registered-user directory
Tasks	GET/POST /api/tasks	Read accessible tasks or create personal/admin-assigned work
Task item	PUT/DELETE /api/tasks/:id	Permission-scoped update and deletion
Notifications	GET /api/notifications	Accessible delivery audit entries
users	MySQL table	Identity, password hash, and admin/user role
tasks	MySQL table	Owner, creator, personal/assigned type, content, status, priority, deadlines
email_notifications	MySQL table	SENT, FAILED, SKIPPED, or SIMULATED delivery records

Database initialization creates missing structures and applies safe upgrades automatically. The dedicated migration files support DBA-managed environments without dropping user data.

ENGINEERING

Security, validation, and reliability

Controls applied across the browser, API, database, email, and scheduler.

Authentication

- bcrypt cost 12 password hashes
- JWT secret must be random and at least 32 characters
- Issuer validation, expiry, and live account lookup
- 401 clears an invalid client session

Authorization

- Admin identity is checked by stored role and configured email
- User directory is admin-only
- Personal rows are owner-only
- Assignees may update only status/completed fields

Input and API

- Parameterized SQL queries
- Strict ID, email, date, enum, length, and boolean validation
- 100 KB JSON limit
- Generic production-safe server errors

Browser and transport

- Explicit CORS origin allowlist
- X-Content-Type-Options, Referrer-Policy, and frame denial
- CSV formula-injection mitigation
- TLS certificate validation enabled by default

Reminders

- Asia/Kolkata timezone configuration
- MySQL advisory lock prevents duplicate scheduler runs
- Only real delivery marks a reminder sent
- Failed messages remain retryable

Operational safety

- Development endpoints hidden unless explicitly enabled
- Email simulation clearly labeled and never treated as delivery
- Graceful API health response when database is unavailable
- No localStorage fallback for task persistence

QUICK START

Run TaskFlow locally

Windows PowerShell setup for the current repository.

1

Install dependencies

npm.cmd ci

npm.cmd --prefix server ci

2

Create local environment

Copy-Item .env.example .env

3

Configure MySQL and JWT

Set DB_HOST, DB_PORT, DB_USER, DB_PASSWORD, DB_NAME, and a 32+ character JWT_SECRET.

4

Confirm the administrator

ADMIN_EMAIL=muralisubbu11@gmail.com

5

Start both applications

npm.cmd run dev:all

6

Open TaskFlow

http://localhost:5173

Health: http://localhost:5000/api/health

QUALITY ASSURANCE

Verification record

Checks completed against the final administrator/status-only permission model.

PASS

ESLint

No lint errors.

PASS

Production build

Vite build completed successfully.

PASS

Server tests

Reminder/date tests passed without sending email.

PASS

Live MySQL migration

Admin role and task_type/role columns verified.

PASS

Authorization test

All assigned-task details blocked for a normal user; status update persisted.

PASS

Browser capture

19 screens captured with zero runtime exceptions.

PASS

Responsive check

Desktop and mobile views showed no page-level horizontal overflow.

PASS

Cleanup

Temporary screenshot users and tasks removed after capture.

The screenshots use temporary demonstration users and tasks plus any pre-existing project data. The demonstration records were deleted after capture; the PDF and source application were retained.